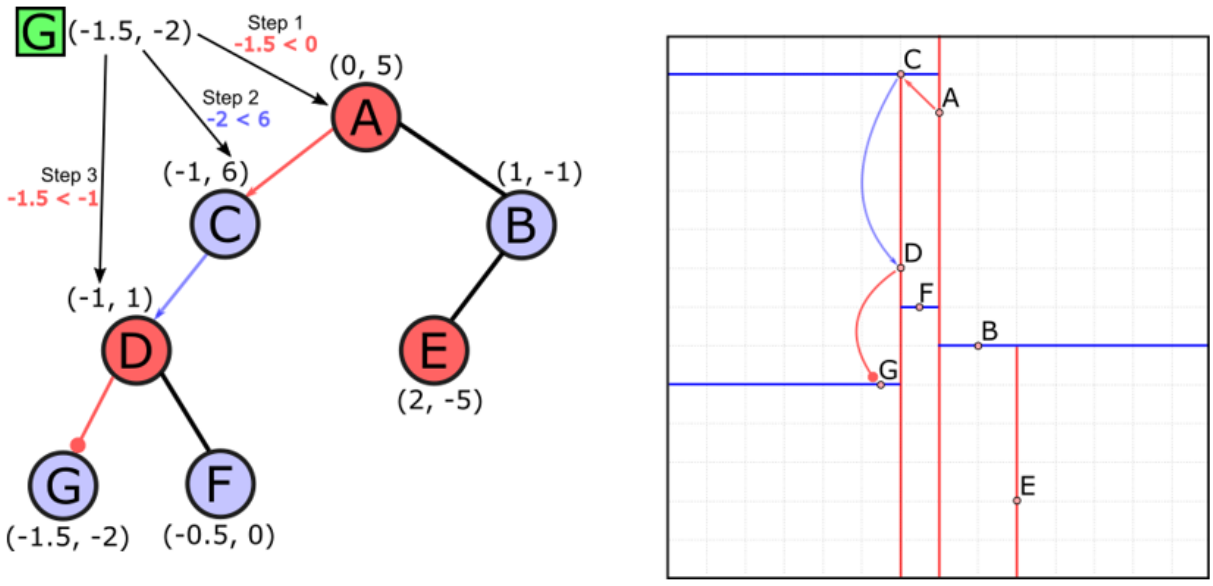


Figure 9.11 Inserting a new point on a k-d tree (2-D). To the k-d tree in figure 9.10 we add a new point `G`: insertion, like for *BST*s, consists of a (unsuccessful) search, after which we can identify the node that will be the parent of the node to add. In case the new point already exists on the tree, your conflicts resolution policy might lead you to ignore the duplicate, or handle it properly.

Notice how, at line #4, we are breaking ties by partitioning coordinates with the same value as current node on its right: this decision might seem of little importance, but we will see that it's a big deal when it comes to deleting nodes.

If we take a look at the stack trace for the example in figure 9.12, we can notice how it is entirely similar to the one for the previous case:

```
insert(A, (2.5, -3))
  insert(B, (-1.5, 2), 1)
```